

How Older Adults Communicate their Technology Problems: Challenges and Design Opportunities

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Abstract

When seeking technology support, the first step is effectively communicating the problem—whether to a person, such as a customer service representative or family member, or a computer program, like a search engine or chatbot. Older adults often face challenges in this process, which can delay or complicate finding accurate solutions. Through a diary study of English-speaking, community-dwelling older adults ($n = 27$, 57 entries), we identify four common articulation issues: verbosity (excessive detail), incompleteness (missing context), over-specification (unnecessary constraints), and under-specification (vagueness). To address these challenges, we examine how foundation models can paraphrase older adults' technology queries to improve solution accuracy. Using a few-shot prompt chaining approach with GPT-4o, we find that AI-assisted paraphrasing significantly improves accuracy: 69% of responses were correct with paraphrased queries vs. 46% with original phrasing. Even in simple Google searches, paraphrased queries yielded 69% accuracy vs. 35% with original queries.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; *Accessibility systems and tools*.

Keywords

older adults, tech support, accessibility, foundation models, question answering

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1 Introduction

Digital applications (apps) and services have become essential for full participation in society, democracy, and the economy. As a result, there is a rising urgency among older adults to effectively use computer technology [14, 15, 30, 50]. Nevertheless, most older adults remain in a *forever-early* adoption stage—using only a few apps, services, and features [20, 33, 34, 66]. This limited technology (tech) use among older adults can stem from different factors, such as a lack of digital awareness [43], a lack of digital confidence [2–4, 11], difficulty learning new tech [6, 27, 46, 49, 52, 57], or difficulty obtaining tech support [46, 51, 52]. For example, during ongoing tech use, older adults often seek tech support when learning something new or troubleshooting a problem. They may ask for help from their social circles (friends and family), community services [28, 39, 46, 51, 52], or customer services [13, 46, 52]—or explore on their own, using help bots [26] or search engines [46, 52]. In any case, the first step is older adults *communicating* the tech issue to a person or a computer program.

The use of computer technology and the process of obtaining tech support among older adults have been widely studied (e.g., [6, 39, 52]). However, fewer studies have explored how older adults *communicate* their tech problems—which is a prerequisite to obtaining any tech support. A recent study examining older users' think-aloud verbalization and speech features found that when encountering an interaction problem with a physical (coffee machine) or digital tech (web and mobile app), older adults tend to use question words like 'what,' 'why,' and 'how' [18]. Another study established a taxonomy of questions older adults ask when encountering an interaction problem with mobile apps [64]: *validation* (asking to verify whether they are doing the right thing, like clicking the right button), *informational* (asking for particular information), *navigational* (asking where a particular user interface function or button is), and *conceptual* (asking about the general workflow or system model that is not directly related to the user interface). Informational questions could be *directed* or *undirected*, either with or without a clear idea about the next action, respectively [64]. Older adults mostly asked validation (38%), directed

informational (23%), and navigational questions (18%). These studies used a think-aloud technique and prescribed well-defined tasks in structured study sessions [18, 64]. What happens when the tasks are either more complicated or less defined and organically occur in older adults' day-to-day lives? Past interview studies have reported that older adults often struggle to search for tech solutions online due to uncertainty or lack of confidence in what to search for. Similarly, tech helpers have reported difficulties in understanding older adults' tech problems, whether assisting them face-to-face or remotely [51, 52]. However, we do not know much about the challenges when older adults communicate their tech problems in unstructured contexts.

Elsewhere, in the aging literature, age-related changes in verbal ability are well documented—for instance, due to a decline in the speed of information processing or a decreased working memory span [9, 22, 32, 41]. For example, our ability to generate the vocabulary that represents action (*action verbal fluency*) declines with age [29]. However, older adults generally demonstrate a vocabulary knowledge advantage over their younger counterparts, which may help compensate for age-related declines in other executive processes [9]. Yet, this advantage in verbal knowledge diminishes when discussing the latest computing technologies, as older adults typically acquire that specific vocabulary much later in life. Furthermore, common age-related cognitive conditions, such as mild cognitive impairment (MCI) or Alzheimer's disease and related dementias (ADRD), are known to negatively impact verbal tasks [17, 42, 58, 61].

Older adults may also struggle to acquire an accurate mental model of digital apps due to a decline in *fluid intelligence* [5]. Fluid intelligence is the ability to perceive relations and correlate, form concepts, reason, and abstract [10, 23]. Mental models are organizing structures that can help people interpret the world [21]. Because acquiring a new mental model requires more cognitive resources than extending an already-acquired mental model [21], older adults find the former more difficult than the latter [21, 44]. However, apps and services that offer similar functions can vary significantly in user interface design and are frequently redesigned. Thus, older adults must acquire newer mental models more frequently than, for example, everyday consumer tech, like an oven or a fridge (although smart home appliances may pose a similar issue). In a recent study, older adults often asked conceptual questions about how a new mobile app worked even when they were familiar with similar apps (e.g., an Amazon app user using the Target app for the first time) [64]. So, any combination of the following factors—a limited vocabulary about computer tech, a decline in verbal fluency, or an inaccurate mental model can make it difficult for older adults to effectively communicate their technology problems.

Helping older adults communicate their tech problems effectively can significantly improve their help-seeking experience—particularly when seeking help independently (self-reliant tech support [52, 53]). It could open up opportunities to leverage the recent advances in user interface (UI) understanding using foundation models [59, 63]. For example, recent studies have explored using large language models (LLMs) to build autonomous agents to answer user screen-related questions [59] or provide interactive tutorials from a pre-existing database [19]. However, these solutions are limited in two ways when considering older adults. First,

these solutions require a well-formed question (e.g., “What’s the rate per night for the Fabulous King Room?” [59]). Second, the question must be asked when the user interacts with the most contextually relevant UI screen, and that contextual information must be directly accessible to the models, e.g., the image of the screen [63], the screen HTML [59], the app-related information, or touch coordinates [19]. However, older adults may not ask clear questions [19] nor know the optimal screen to ask questions during an interaction [52]. Furthermore, some questions may not be related to any particular UI screen or feature.

In this paper, we examine how English-speaking older adults communicate technology problems in unstructured contexts and explore how foundation models can make those queries easier to solve using computer programs. Our work contributes: (1) a characterization of older adults' tech queries occurring organically in day-to-day life; (2) a *few-shot prompt chaining* approach with foundation models to paraphrase those queries to improve the likelihood of finding quick and accurate solutions; and (3) an evaluation of the proposed approach with a large language model, GPT-4o [45], and a foundation action model, OS-ATLAS [63].

2 Communicating Tech Problems

Older adults must communicate tech problems to find solutions. The solutions can be delayed or inaccurate if their query can not be readily searched online or understood by a tech helper. We set out to explore how older adults communicate their (computer) tech problems in *unstructured* contexts, i.e., not within a research study session where certain tasks, apps, features, or services could be prescribed. We examine the following research question: *How do English-speaking, community-dwelling older adults ask tech questions that naturally occur during their daily lives?*

To our knowledge, no prior work has examined this particular research question. A recent study with Chinese older adults reported collecting questions during simulated help-seeking interview sessions [19]. During those interviews, researchers identified a set of functions or problems (e.g., “Have you ever encountered a situation where you did not want to receive messages from your friends?”, p. 7, [19]) in commonly used apps to simulate help-seeking behavior. Other interview studies have also indicated that how older adults communicate their tech problems, whether synchronous or asynchronous, may be problematic [51, 52]. However, there is a lack of empirical data to characterize the nature of these problems—before solutions can be explored.

2.1 Methods

Given the different settings in which tech questions could be asked, we focused on one-to-one asynchronous messaging with a tech helper. We conducted an *online diary study*, which was approved by a university-wide institutional review board. We recruited people 60 years or older who currently owned and used a mobile device (e.g., smartphone or tablet) at least once a week and were proficient in speaking English. We chose to focus on mobile devices, as the latest research suggests they are the most common computer devices owned and used by older adults [50]. Participants were recruited through various methods, including flyers, community announcements, email lists, social media platforms, and

Table 1: Key characteristics of older adults' tech queries.

Characteristic	Count	Example
Verbosity	9/57	<i>"These are the START SCREENS screenshots on [name]'s computer. One for searching, the usual purpose for booting up, and one for Click Bait. My computer combines the two and I do spend some wasted time clicking on the icons. Any way to combine them. Or better, just dispense with the clickbait. And BTW, the favorites are not necessarily the same. She hasn't complained about hitting Facebook and getting my Facebook account – Lately."</i> —80, man
Over-specification	12/57	<i>"Argh.. I have had this problem for a while and don't even know how or why it happened. I want to watch my wolf utubes with my phone or iPad in the horizontal position. All of a sudden it won't turn when I turn my phone. What happened?!! How do I fix it?!!"</i> —66, woman
Under-specification	13/57	<i>"Hi dear, I'm trying to watch a cooking video on YouTube, but I can't understand what they're saying. Can you help me?"</i> —62, woman
Incompleteness	24/57	<i>"I come back to phone and it's blown up so huge I can't move it. [name] helped me resolve it by just clicking on it. There's gonna be one every day."</i> —61, woman

word-of-mouth. The online diary study lasted up to eight weeks, but participants could leave the study at any time. No monetary compensation was provided for participation. After initial contact, a researcher (first author) met with participants to introduce the research study. Participants were asked to message the research team via their preferred form: email, text, or WhatsApp. They were invited to ask about any tech problems related to mobile devices. Participants could communicate via text, screenshots, voice messages, or video recordings of their screens. They were asked not to share any personal information at any point during the study. Automatic reminders were sent every two days to encourage reporting. As an incentive, a member of the research team provided help in resolving the reported problem. All communication, except the initial introduction, occurred via asynchronous messaging.

Midway through the study, after a low response rate (one entry per week, on average), we uncovered (via interviews) that older adults often sent their problems to family members (unofficial tech helpers) first and to the research team later—mostly if the problem remained unsolved. To address the low participation, we next recruited older adults indirectly via their tech helpers (18+) who assisted older adults with their tech problems. Tech helpers were asked to document when they helped older adults asynchronously and share their message exchanges with us. Helpers also shared the tech problem in their own words and whether or not it was successfully resolved. Note that this was a diary study, and not all participants remained engaged for the full eight weeks. Furthermore, we captured only technology-related queries that arose naturally during older adults' day-to-day activities. Participant retention in longitudinal studies, particularly those involving older adults, is often challenging due to factors such as motivation, cognitive load, and competing responsibilities [1, 16, 56].

2.2 Results

2.2.1 Participants. We collected diary entries from 27 older adults (13 women, $Mdn_{age} = 66$, one participant declined to respond). 17/27 were indirectly recruited via their tech helpers. Only one of the ten tech helpers assisted an older adult who also participated in our study. One participant asked a maximum of ten queries

over eight weeks. All queries were unique. No one asked about the same problem more than once. Participants owned various devices, including iPhones, iPads, Android phones, and Amazon Fire tablets.

2.2.2 Technology Queries. We collected 57 queries in text format, nine of which included a screenshot. No voice or video recordings were provided. The tech queries ranged across 34 unique apps. Two of those problems remained unresolved due to participant attrition. The resulting corpus consisted of the original tech queries and all text messages between the tech helper and the older adult during follow-up. The original tech queries ranged from 5 to 223 words per query ($Mdn = 65$, $IQR = 27$). We used the taxonomy from [64] to categorize the 57 queries. The first and last authors coded them independently (Cohen's $\kappa = .72$). Upon resolution of any discrepancies by the last author, there were four validation, 14 directed informational, 23 undirected informational, five navigational, and 11 conceptual questions.

To better understand any additional characteristics of the original tech queries, a reflexive thematic analysis was conducted [7, 8]. The first author analyzed the queries inductively using open coding, referencing the entire conversation as needed. Then, the codes were discussed and iterated in group data analysis sessions until key themes were identified. To identify the open codes, we grounded our analysis in the aging literature about verbal knowledge, verbal fluency, and age-related changes in fluid and crystallized intelligence. Following data analysis, we identified four key characteristics of older adults' tech queries: *verbosity*, *over-specification*, *under-specification*, and *incompleteness* (Table 1).

The tech queries could be rambling and include unnecessary details, thereby hindering a swift understanding of the problem (e.g., *"My wife [name] has been advised by trusted source that...Actually, it sounds like something we should both do from time to time...I'm also confused..."*—70, man). On the other hand, some queries were brief but filled with esoteric details that made it challenging to understand the core issue (e.g., *"Need mobile recharge statement for reimbursement but not able to download from app. My son is out of station and he asked for recharge statement for his company for reimbursement"*—60, woman). Tech queries could also lack minute

but essential contextual details, like canceling a scheduled or active rideshare, or be a combination of multiple queries missing scoping details pertaining to each sub-query (e.g., “*I’m really having trouble with my Kindle app. It’s been acting weird - the books I read daily aren’t syncing between my phone and tablet, and I can’t download any new ones. Also, the text is too small, and I have no idea how to fix it.*” —60, man). Is the smaller text problem only related to the tablet, phone, or newly downloaded books? Finally, the most common characteristic of these tech queries was a lack of necessary contextual information, like what device, operating system, or app they had the problem with. Moreover, older adults occasionally used unusual vocabulary that required further clarification to understand, e.g., ‘*power going off*’ to mean phone getting locked, ‘*screen going dark*’ to mean dark theme, and ‘*tongue*’ to mean voice commands. Given these challenges when communicating tech problems among older adults, we wanted to explore ways to reformulate the original tech query to make it easier and quicker to resolve.

3 Reformulating Tech Problems

After examining the different characteristics of older adults’ tech queries, we set out to reformulate them so that they can be resolved independently, for example, by searching online or asking a chatbot. When designing a solution, we chose to address two characteristics of our dataset: (1) 9/57 entries included a screenshot of the relevant UI, implying screen-related information is a rarity, and (2) a lack of essential contextual information that the tech helpers had to elicit via follow up questions in order to resolve the tech problem. So, unlike existing question-answering solutions that assume pertinent UI information is always available [19, 59, 63], can we proactively obtain contextual information by asking older adults the right questions? To that aim, we explored a *prompt-chaining* approach with foundation models, first, to generate contextual questions, next to paraphrase the original tech query, and then, finally, to generate a solution (Figure 1).

3.1 Methods

3.1.1 Model Selection. We selected two foundation models, GPT-4o [45] and OS-ATLAS [63], to examine our approach. OS-ATLAS is a foundational GUI action model designed for GUI grounding and out-of-distribution agentic tasks. Our decision to include OS-ATLAS was driven by its capabilities in GUI grounding and its potential for integration into systems without relying on external APIs. GPT-4o was chosen as the general-purpose language model due to its strong generalization and versatility [31, 37, 62]. Compared to other large language models (LLMs) such as Gemini, LLaMA, and Claude; GPT-4o demonstrates superior reasoning ability, text causality, and overall response comprehensiveness and accuracy [38].

3.1.2 Dataset. A portion of the corpus collected through our online diary study was used as a dataset for prompt engineering and model evaluation. We included only tech queries that were meaningful without the accompanying screenshots and excluded those that remained unresolved after follow-ups with tech helpers. For each of the remaining 51 tech queries, two researchers independently reviewed the queries (along with follow-up conversations, when available) and identified the underlying tech problem (Cohen’s $\kappa = .89$).

The final dataset consisted of 51 queries, their associated follow-up conversations, and expert annotations identifying the tech problem.

3.1.3 Prompt Design. We developed confidence-aware prompts for GPT-4o and OS-ATLAS using an iterative few-shot approach. Iterations were conducted independently for each model to ensure optimal performance tailored to their respective architectures. While the final prompts for GPT-4o and OS-ATLAS differed due to these iterations (Figures 2 and 3), the few-shot examples remained consistent across both models to enable a fair comparison. Three of the 51 original tech queries were used as examples during the prompt design phase (Figure 1). Each prompt was structured to guide the models through the following tasks: (1) analyze the original tech query and identify missing essential contextual details; (2) generate up to three follow-up questions to elicit that information; (3) paraphrase the original query into a concise, complete format (one or more questions) suitable for direct input into a search engine like Google; and (4) provide a single, confident solution ($\geq 80\%$ confidence for OS-ATLAS and $\geq 90\%$ for GPT-4o) to the tech problem described in the query.

Throughout the iterations, we tested the prompts using representative examples that covered diverse query types, including step-by-step instructions, explanations of technology concepts, and handling of impossible asks. Feedback from these tests was used to refine the prompts, addressing issues such as redundancy in follow-up questions, omission of important details in the paraphrased question, and the use of overly technical terminology in the solution. Instructions were added to enable the models to break down complex requests into multiple paraphrased questions. Both models were explicitly directed to prioritize clarity and simplicity in their solutions, avoiding multiple alternatives to enhance usability for older adults.

To mitigate hallucinations, we set a confidence threshold for response generation. Since LLM confidence is not a direct statistical measure but rather an internal estimate of certainty [25, 36], we heuristically determined a 90% threshold for GPT-4o. A lower threshold (80%) was set for OS-ATLAS due to its more conservative confidence calibration. While not a direct measure of accuracy, these thresholds helped ensure that responses were generated only when the model exhibited sufficient comprehension, thereby improving reliability. The iterative refinement process ensured that each model’s prompt achieved strong performance while aligning with our design objectives. The final prompts are available in the Appendix (Figures 2, 3, and 4).

3.1.4 Model Workflow. The GPT-4o model was accessed via the OpenAI API, while the OS-ATLAS model ran locally. When evaluating the 48 original tech queries, we followed this workflow.

- (1) The original query is input into the model, which is then instructed to generate follow-up contextual questions.
- (2) Responses to these generated questions, i.e., contextual information, are provided to the model from our corpus collected through the diary study. If no relevant response is available, the model receives “I do not know” as input. Notably, our corpus includes not only the queries but also the conversations between tech helpers and older adults leading up

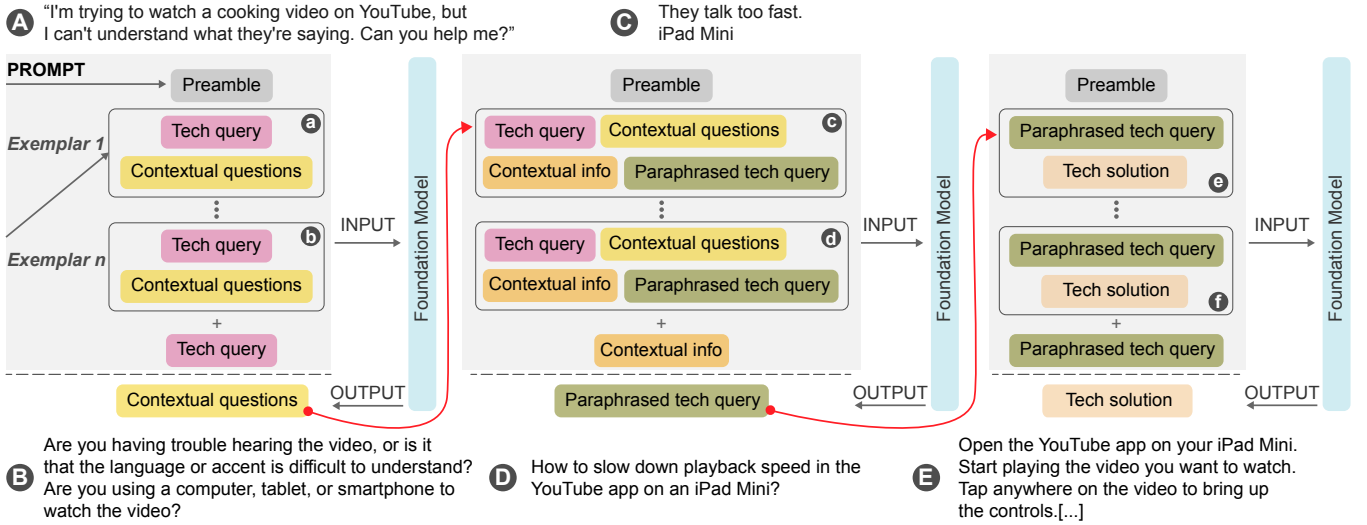


Figure 1: Our *few-shot prompt chaining* approach with foundation models operates as follows: First, a tech query from an older adult is input into the model (A), which is then instructed to generate follow-up contextual questions (B). The model is provided with a few examples (a, b) for guidance. Older adults would then answer these questions to supply contextual information (C). Next, the model takes the original query, contextual questions, and provided contextual information as inputs to generate a paraphrased query (D), with additional examples (c, d) included for reference. Finally, using another set of examples (e, f), the model is asked to generate a solution to the paraphrased tech query (E).

to problem resolution. Most resolutions required eliciting additional contextual information.

- (3) The model is subsequently instructed to paraphrase the original query into a concise, complete format suitable for a simple Google search.
- (4) Finally, the model is asked to provide a single solution to the paraphrased query. Solutions may include step-by-step instructions, conceptual explanations, or a brief rationale for why the query could not be resolved.

3.2 Results

Model outputs were evaluated for *paraphrasing* and *solution-giving* effectiveness. Paraphrasing effectiveness was assessed based on expert judgment and categorized into three levels: correct, partially correct, and incorrect (e.g., see Figures 5 and 6). Partially correct paraphrasing either omitted essential parts of the original query or introduced hallucinated details. Two experts (the first and second authors) independently rated the paraphrasing effectiveness of both models, achieving inter-rater agreement (Cohen's κ) of .75 for OS-ATLAS and .85 for GPT-4o. Disagreements were discussed and resolved before final categorization (Table 2). We did not evaluate the paraphrased queries with older adults for two reasons. First, the goal of paraphrasing was to improve the likelihood of obtaining a timely and accurate solution, rather than to assess how older adults perceived the paraphrased queries. Second, the same challenges that led older adults to generate problematic original queries would likely prevent them from accurately judging the effectiveness of the paraphrased queries [40, 60].

Solution-giving effectiveness was assessed based on the perceived accuracy of the solution, categorized into three levels: correct, partially correct, and incorrect. Examples illustrating these levels are provided in Figures 7 and 8. The ground truth for this evaluation was available because all tech problems had been resolved during the diary study with older adults. The expert solution for each technology query was created by the first author, who extracted the correct steps from the diary study conversations and structured them following the same instructions given to the models in the prompts. A comparison of the two models' effectiveness is presented in Table 3. In the Google search condition, all correct solutions appeared within the first five results, excluding AI-generated content. Only the top 20 Google search results were reviewed to identify a solution. Both models responded with "I do not know" to three queries, one of which was common to both.

Paraphrasing was quantitatively assessed using ParaScorer [54] with two types of metrics: *reference-based*, which compares model outputs to expert-generated reference paraphrases, and *reference-free*, which evaluates paraphrase quality without a predefined reference by assessing factors such as semantic similarity and fluency. While both semantic similarity and lexical divergence are important for reference-free evaluation, high lexical divergence is often penalized. GPT-4o (.95) outperformed OS-ATLAS (.85) in reference-based evaluation, whereas OS-ATLAS (.70) outperformed GPT-4o (.60) in reference-free evaluation. To contextualize these metrics, rephrasing an older adult's original tech query often requires incorporating additional contextual information that was not present in the original query, leading to higher lexical divergence. Thus, despite its lower score in reference-free evaluation, GPT-4o better aligns with our objective. Moreover, its strong performance on reference-based

Table 2: Expert evaluation of paraphrasing effectiveness (%)

	GPT-4o	OS-Atlas
Correct	68.8	12.6
Partially correct	25	43.7
Incorrect	6.2	43.7

metrics indicates a closer alignment with expert-generated solutions. We also evaluated paraphrasing performance against expert paraphrases using general NLP-based metrics, including contextual similarity (BERTScore [65]), machine translation quality with a reference translation (BLEU [47]), summary quality (ROUGE-L [35]), and semantic similarity (Cosine similarity [48]). Table 4 presents the similarities between model-generated and expert-generated paraphrases, indicating that GPT-4o closely aligns with expert solutions. These metrics were also used to evaluate model-generated solutions against expert solutions (Table 5). In BLEU and ROUGE, GPT-4o achieved scores of .62 and .72, while OS-ATLAS scored .04 and .30, respectively. These metrics assess n-gram similarities, which are essential for evaluating the correctness of model-generated solutions. When the solution involves sequential steps, accurately matching the sequence is equally important for achieving a correct and complete solution.

4 Discussion

Tech ownership among older adults is rapidly approaching parity with younger adults, yet obtaining tech support remains a challenge [53]. Older adults commonly seek assistance from people [24, 39] or computer programs [12, 19, 46, 55]. However, how they present their tech problems can significantly impact the effectiveness of the support they receive. In this paper, we analyzed how English-speaking, community-dwelling older adults ask tech-related questions in unstructured contexts. Our online diary study revealed that tech helpers need to frequently ask follow-up questions to gather additional contextual information before they can understand the problem and provide a solution. Current support systems, however, assume that this contextual information is readily available, often in forms such as screen HTML [59] or touch coordinates [19]. Moreover, these systems primarily target queries that are either directed

Table 3: Expert evaluation of solution effectiveness (%)

Paraphrasing by	None	GPT-4o	OS-Atlas	OS-Atlas	GPT-4o
Solution by	Google	Google	Google	OS-Atlas	GPT-4o
Correct	35.4	68.8	35.4	10.4	68.8
Partially correct	2.1	22.9	10.4	10.4	14.6
Incorrect	62.5	8.3	50	79.2	16.7

informational (e.g., “What’s the title of the newest mail?” [59]) or navigational (e.g., “I want to collect this” [19]). In contrast, our dataset shows that only 25% of older adults’ queries were directed informational, and just 9% were navigational. Additionally, older adults’ tech queries exhibited characteristics that could hinder quick and accurate resolution (Table 1), findings that align with recent research on how Chinese older adults articulate tech queries [19]. Although our dataset is limited in size, our findings highlight the importance of designing tech support tools that help older adults better formulate their tech problems before seeking a solution.

We proposed a new approach to reformulating older adults’ tech queries that mimics the workflow of a tech helper—asking contextual questions to understand the problem before attempting to solve it. Instead of relying on human assistance, we used foundation models to generate these contextual questions. Our few-shot prompt-chaining approach outperformed the baseline: when paraphrased by GPT-4o, a correct solution appeared within the top five Google search results 69% of the time, compared to 35% with the original queries from older adults (Table 3). Across all metrics, GPT-4o outperformed OS-ATLAS, despite the latter being trained on graphical user interface (GUI) screenshots while GPT-4o was trained on broad textual data. This difference is likely because OS-ATLAS is optimized for basic GUI actions, such as clicking and scrolling, but struggles with complex workflows. In contrast, foundation models like GPT are pretrained on large datasets containing broad general knowledge, enabling strong performance on various tasks without fine-tuning. This allowed us to explore our approach using few-shot prompting, even with a small dataset from our diary study. That said, OS-ATLAS may have performed better if the textual queries had been accompanied by relevant screenshots. However, older adults in our diary study rarely included screenshots when describing their tech problems.

We evaluated model outputs against expert data but did not assess the quality of the contextual questions—specifically, whether the models effectively gathered the necessary context to improve problem resolution or merely generated follow-ups without meaningful contribution. This remains an area for future research. Additionally, we did not evaluate whether older adults could understand the model-generated follow-up questions and provide the necessary contextual information. Further research is also needed to assess how efficiently and effectively older adults can follow model-generated solutions—the ultimate goal of this work. Another key consideration is how younger people, who frequently assist older adults with technology, perceive the paraphrased tech queries. Do these reformulated queries make their role easier? Future studies should explore this to ensure model-generated outputs align with

Table 4: Quantitative evaluation of query paraphrasing.

Model	BERTScore	BLEU	ROUGE-L	Cosine
GPT-4o	.98	.75	.87	.88
OS-ATLAS	.92	.09	.40	.36

Table 5: Quantitative evaluation of solutions generated.

Model	BERTScore	BLEU	ROUGE-L	Cosine
GPT-4o	.94	.62	.72	.79
OS-ATLAS	.87	.04	.30	.38

real-world needs. Finally, our study focused on tech communication in asynchronous contexts. Other settings, such as face-to-face interactions with family members or customer service representatives, may exhibit similar or additional challenges that influence how older adults describe their tech problems. Future work should examine these alternative contexts to develop more comprehensive support strategies.

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A Appendix

A.1 GPT-4o Prompts

An older adult has described the following problem:

{Original Query}

Your task is to assist them in solving their issue. For that, the first step is:

Determine if the problem description contains all the necessary information to resolve the issue. If all information is provided: Print that you are ready to proceed to the next step. If additional information is needed: Ask up to three specific and concise follow-up questions to clarify any missing details. Only ask questions that are essential to solving the problem. Avoid unnecessary or overly technical questions. You don't need to ask 3 questions, you can ask less. Provide only the questions without additional text.

Example 1:

User Query: "On my desktop always have the option of reverting to a prior date's settings or configurations. Is there a similar feature for Android phones?"

Model Response: No questions, I can solve it.

Example 2:

User Query: "Argh.. I have had this problem for a while and don't even know how or why it happened. I want to watch my wolf utubes with my phone or iPad in the horizontal position. All of a sudden it won't turn when I turn my phone. What happened?!! How do I fix it?!!"

Model Response: 1. What type of phone do you have? (For example, iPhone or Android) 2. Have you noticed this happening with applications other than YouTube?

Example 3:

User Query: "It seems that "System Apps" can't be deleted, but can be stripped of updates and remain functioning. Is this my insurance against deletion?"

Model Response: 1. What is the device type? (For example, Laptop, Tablet, or Phone) 2. Do you know if it is an Apple product or not?

--End of examples--

Figure 2: The prompt used for generating contextual questions.

Now, I will send you the user's answers to the questions you asked in the previous turn:
{Answers to the follow-up questions, if available}

After reviewing the answers to the follow-up questions, rephrase the question in a way that is clear enough for the user to copy and paste into a Google Search Bar and be able to find the answer. If you think there are multiple questions with different solutions being asked, separately rephrase them. If you are less than 90% confident in your understanding of the user's intent and problem, just say, 'I don't know.' Provide only the rephrased questions without additional text.

Example 1:

Technology question: On my desktop always have the option of reverting to a prior date's settings or configurations. Is there a similar feature for Android phones?

Model Response: No questions, I can solve it. Can I revert back my settings and configurations to a prior date on an Android device?

Example 2:

Technology question: Argh.. I have had this problem for a while and don't even know how or why it happened. I want to watch my wolf utubes with my phone or iPad in the horizontal position. All of a sudden it won't turn when I turn my phone. What happened?!! How do I fix it?!!

Model Response: 1. What type of phone do you have? (For example, iPhone or Android)

2. Have you noticed this happening with applications other than YouTube?

User: 1. iPhone 2. I don't know

Model Response: How to turn off Portrait Orientation Lock on an iPhone?

Example 3:

Technology question: "It seems that 'System Apps' can't be deleted, but can be stripped of updates and remain functioning. Is this my insurance against deletion?"

Model Response: 1. What is the device type? (For example, Laptop, Tablet, or Phone) 2. Do you know if it is an Apple product or not?

User: It's a Samsung phone."

Model Response: Can I delete system apps on an Android phone accidentally?

--End of examples--

Figure 3: The prompt used for paraphrasing a tech query.

{Paraphrased Question generated by the model}

Now, provide step-by-step instructions on how to solve the tech issue(s) in a language that is understandable for older adults. Keep each step short, and the instructions concise. If you are less than 90% confident that the solution would work for the user, just say, 'I don't know.' Providing incorrect solutions will frustrate our users. If the user's request is impossible, say, 'You cannot do that' with a brief explanation of why. If no action from the user is required, say, 'No action needed' with a brief explanation of why. Do not offer alternative solutions. Only give the one solution you are most confident in. Do not ask any more questions from the user.

Example 1:

Paraphrased question: Can I revert back my settings and configurations to a prior date on an Android device?

Model response: Unfortunately, that is not possible on an Android phone.

Example 2:

Paraphrased question: How to turn off Portrait Orientation Lock on an iPhone?

Model response: Here is step-by-step instructions:

1. Swipe down from the top-right corner of your screen to open Control Center.
2. Tap the button that has a lock inside and make sure that it's off.
3. Go back to your video and turn your iPhone sideways.

Example 3:

Paraphrased question: Can I delete system apps on an Android phone accidentally?

Model response: You can't delete system apps that came pre-installed on your Android phone.

--End of examples--

Figure 4: The prompt used for generating a tech solution.

User: When I go into Wi-Fi on Settings on my iPad, it states 'Privacy Warning' under the network I'm connected to. It doesn't state that on my iPhone though. Do you know what that means and if I need to do something about it?

Model: Is the Wi-Fi network you're connected to at home, work, or a public place?

User: Home

Model: Why does my home Wi-Fi network show a "Privacy Warning" on my iPad but not on my iPhone?

Figure 5: Example of a correct paraphrasing.

User: Occasionally when working on my laptop the screen I'm working on will just slide off the right. And I'm left with just my desktop. If it was a document I was working then I know that I can go to the Microsoft Word icon on my toolbar, click it, and the homescreen will "shift" again and my document will reappear. But I don't know what I'm doing that would have caused the document to slide off screen to begin with. This doesn't happen frequently, just occasionally as I said. And frustrates me when it does.

Model: 1. What type of laptop do you have? (For example, Dell, HP, or Lenovo)

2. What operating system is installed on your laptop? (For example, Windows, macOS, or Linux)

3. Are you using a specific application that causes the screen to slide off the right?

User: 1. MacBook 2. MacOS Ventura 3. Word, but it can happen with other apps too.

Model: Why does my laptop screen slide off the right occasionally and how can I fix it?

Correct paraphrasing: Why does my current window slide off-screen on macOS Ventura, and how do I get it back?

Figure 6: Example of an incorrect paraphrasing.

Paraphrased query: How do I make everything on my iPhone, including app icons and text, appear larger?

Model: Here are step-by-step instructions:

1. Open the Settings app on your iPhone.
 2. Scroll down and tap Display & Brightness.
 3. Tap Text Size to make text larger. Use the slider to adjust the size.
 4. If you need even larger text, go back and tap Accessibility.
 5. Select Display & Text Size and then tap Larger Text.
 6. Turn on Larger Accessibility Sizes and move the slider to make everything bigger.
 7. To make icons and buttons larger, go back to Accessibility and tap Zoom.
 8. Turn on Zoom and follow the on-screen instructions to zoom in and out.
- This should make apps, text, and icons easier to see.

Figure 7: Example of a correct tech solution.

Paraphrased query: How do I cancel a scheduled Lyft ride?

Model: Here are step-by-step instructions:

1. Open the Lyft app on your phone.
 2. Tap on the menu icon (three lines) in the top-left corner.
 3. Select "Ride History" from the menu.
 4. Find the scheduled ride you want to cancel.
 5. Tap on the ride, and then tap "Cancel Ride."
 6. Confirm your cancellation when prompted.
- That's it! Your ride should now be canceled.

Correct solution: Here are step-by-step instructions:

1. Open the Lyft app
2. Tap "Trips" on the bottom of the screen.
3. Locate the trip you want to cancel by scrolling through the trips if necessary.
4. Next to the trip details, find the "Cancel Trip" button.
5. Tap on "Cancel Trip."
6. Confirm your cancellation if prompted.

Figure 8: Example of an incorrect tech solution.